

Controller-in-the-Loop Curation of Kinematic Humanoid Motion

CS 348K Project

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Abstract

Kinematic humanoid motion generators such as MotionBricks produce diverse Unitree G1 reference trajectories, but a kinematically plausible reference is not necessarily executable by a physics-based controller. We study this gap as an acceptance and curation problem. Given a fixed MotionBricks candidate pool, we compute pre-controller diagnostics, evaluate selected references with the native SONIC G1 tracking stack, train a trajectory-level qpos acceptance model from native rollout labels, and audit rendered videos frame-by-frame from MP4 pixels. On a broad 13-mode, 104-identity candidate pool, a hand-coded gated pre-controller selector improves strict native SONIC pass rate from 70/104 for the deterministic baseline to 78/104. A learned temporal qpos classifier predicts strict native acceptance better than scalar diagnostics in grouped cross-validation (AUC 0.864 versus 0.782 for the best scalar), and its prospective all-candidate selections reach 76/104 strict passes. The learned selector improves over the deterministic baseline, but does not beat the hand-coded gate. Its strongest behavior is abstention: thresholding the learned score at 0.5 removes all 16 unsupported crawling identities and keeps 76 strict passes over 88 accepted identities. A follow-up hybrid queue closes the missing native labels and reaches 74/88 strict accepted identities; all 9 newly evaluated hybrid videos pass visual pixel audit, but two idle clips fail strict root-drift criteria. The final claim is therefore not that MotionBricks is fine-tuned or physically solved. The contribution is an honest systems study: native controller labels and visual audits expose failure modes, hand-coded gates are currently strongest, and learned acceptance is useful for triage and abstention before expensive controller rollout.

1 Motivation

MotionBricks is a fast kinematic motion generator with modular latent motion primitives and real-time controllability [4]. Its physical awareness limitation is exactly the problem a robotics user sees in practice: a generated reference can look plausible in pose space yet exceed controller, contact, root-height, or balance limits. Recent humanoid work increasingly treats tracking policies and physical feedback as the arbiter of motion quality: SONIC scales humanoid motion tracking with large motion data and compute [2]; GMT and SMAP study broad whole-body tracking/adaptation [3, 1]; RLPF uses physical feedback to align large motion models with humanoid control [5]. Our project asks a smaller question that can be tested locally: can we curate a fixed MotionBricks candidate pool into more controller-executable references, and can we tell when the selector should abstain?

This framing deliberately avoids an overclaim. We do not fine-tune MotionBricks. We do not train a new tracking policy. We do not prove real-robot safety. Instead, we make the generator/control gap measurable with native SONIC rollouts, learned acceptance prediction, hard gates, and video audits.

2 Pipeline

Candidate pool. The final broad run uses 13 exposed MotionBricks G1 modes, 8 seeds, and stochastic candidate sampling, producing 832 candidates over 104 (mode, seed) identities. The modes include idle, upright walking/style variants, and hand/elbow crawling stress tests. Each candidate is exported as a Unitree G1 qpos trajectory and round-tripped to SONIC reference CSVs; the export audit reports max qpos error 3.33×10^{-16} , so the main failures are not explained by a broad joint-order conversion bug.

Pre-controller diagnostics. For every candidate we compute inverse-dynamics demand, root-height sanity, contact artifacts, self-contact, non-foot floor contact, support proxies, foot skate, velocity, acceleration, and jerk. These are not final physical proof: they are cheap pre-controller triage features.

Native controller labels. Selected references are run through the native SONIC G1 release protocol with the robot released before tracking. A rollout is counted as *strict* if it does not fall, has mean joint RMSE ≤ 0.20 rad, and has mean root XY error ≤ 1.5 m. We report survival separately from strict pass because several clips survive but drift too far.

Learned acceptance model. A temporal qpos CNN is trained on 405 native rollout labels from the broad run. Splits are grouped by identity so duplicate selector choices from the same mode/seed do not leak across folds. The model predicts strict native acceptance from the reference trajectory alone; it is an acceptance model, not a generator.

Visual audit. Numbers are not enough for motion. We rerender native rollouts with camera tracking and contact markers, then run a frame-level MP4 audit that segments the red actual robot and white reference robot in every frame. The audit flags missing bodies, large visual separation, fallen-looking body aspect, and sudden visual jumps. We use this as a systematic review aid, not as a replacement for human validation.

3 Results

3.1 Hand-coded gating is the current strongest full-coverage selector

The broad13 native SONIC evaluation completes 405 selected references. The main identity-level comparison is:

Table 1: Broad13 native SONIC strict pass rates.

Selector	All	Upright	Idle	Crawling
Deterministic baseline	70/104	63/80	7/8	0/16
Best pre-controller score	73/104	–	–	–
Lowest inverse-dynamics risk	72/104	–	–	–
Hand-coded gated selector	78/104	71/80	7/8	0/5 selected
Native oracle over tested choices	88/104	–	–	–

The useful effect is modest but real: hard root/contact gates plus pre-controller scoring rescue some brittle upright references. The gap to the native oracle shows that better selection is possible, but scalar gates are not enough.

3.2 Learned acceptance predicts better than scalar diagnostics

The temporal qpos acceptance model reaches grouped cross-validated AUC 0.864 and average precision 0.917. The best scalar diagnostic is root-z minimum at AUC 0.782; pre-controller score reaches AUC 0.752. This supports the idea that native acceptability is trajectory-structured, not fully captured by a single root/contact/risk scalar.

Table 2: Strict native-acceptance prediction on broad13 rollout labels.

Model / diagnostic	AUC	Average precision
Temporal qpos CNN	0.864	0.917
Root-z minimum	0.782	0.865
Low-root frame percentage	0.768	0.849
Pre-controller score	0.752	0.845
Contact artifact score	0.741	0.839
Full inverse-dynamics risk	0.737	0.845
Torque-limit ratio	0.719	0.832

3.3 Learned selection improves over baseline but not over the gate

We score all 832 broad13 candidates with the learned model and export the best candidate per identity. Native SONIC evaluates all 104 learned selections:

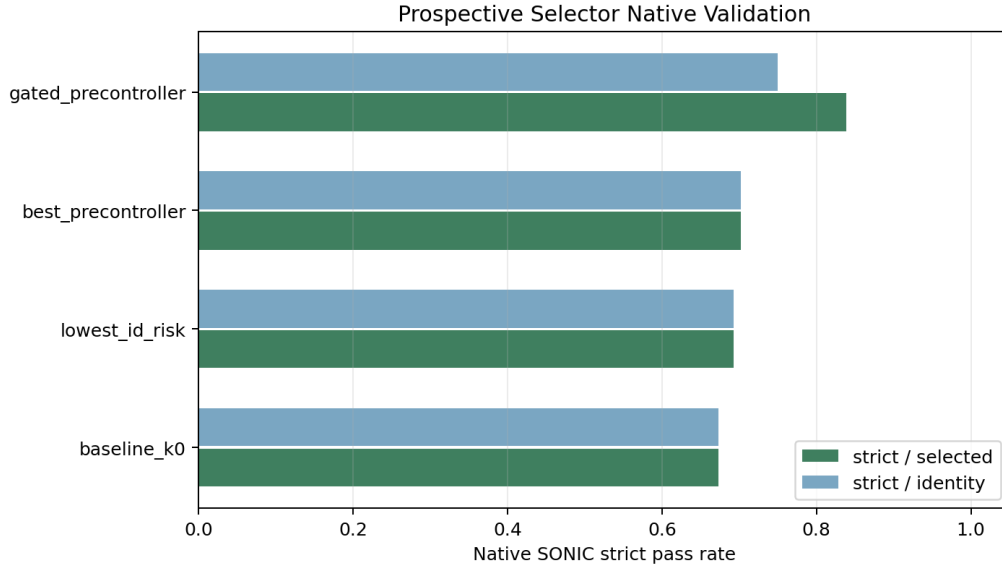


Figure 1: Broad13 selector comparison. The hand-coded gated pre-controller selector is the strongest full-coverage result, while the native oracle shows remaining headroom.

Table 3: Prospective learned-selector native rollout.

Selector	All strict	Upright strict	Idle strict	Crawling strict
Deterministic baseline	70/104	63/80	7/8	0/16
Hand-coded gated selector	78/104	71/80	7/8	0/5 selected
Learned acceptance selector	76/104	68/80	8/8	0/16

This is not a clean deployment split: the learned model selects from the same broad13 candidate pool used to train labels. Of its 104 selected candidates, 46 had already been native-evaluated in the broad selector study, while 58 were newly evaluated here. The newly evaluated subset is 37/58 strict, including 36/42 upright strict and 0/15 newly evaluated crawling. Thus the learned selector is promising, but not a new state-of-the-art result.

3.4 Abstention is the strongest learned behavior

The learned score strongly separates unsupported crawling in this run. A score threshold of 0.5 removes all 16 crawling identities and keeps all 76 strict passes:

Table 4: Learned-score abstention on the 104 learned selections.

Min score	Accepted identities	Strict	Falls	Crawling accepted
0.00	104	76/104	22	16
0.50	88	76/88	6	0
0.80	88	76/88	6	0
0.85	77	65/77	6	0
0.90	72	62/72	6	0

We then build a first all-candidate hybrid queue: reject crawling and low-root references, require the root/contact sanity gate, then rank by learned score. It accepts 88 identities and rejects 16. After evaluating the 9 selected references not already covered by the learned rollout, the closed-label hybrid queue reaches 74/88 strict accepted identities: 68/80 upright and 6/8 idle. All 9 newly evaluated hybrid videos survive and pass visual audit, but two idle clips fail strict because root XY drift exceeds 1.5m. This reveals the next missing feature: idle/root-drift sanity must be part of the selector.

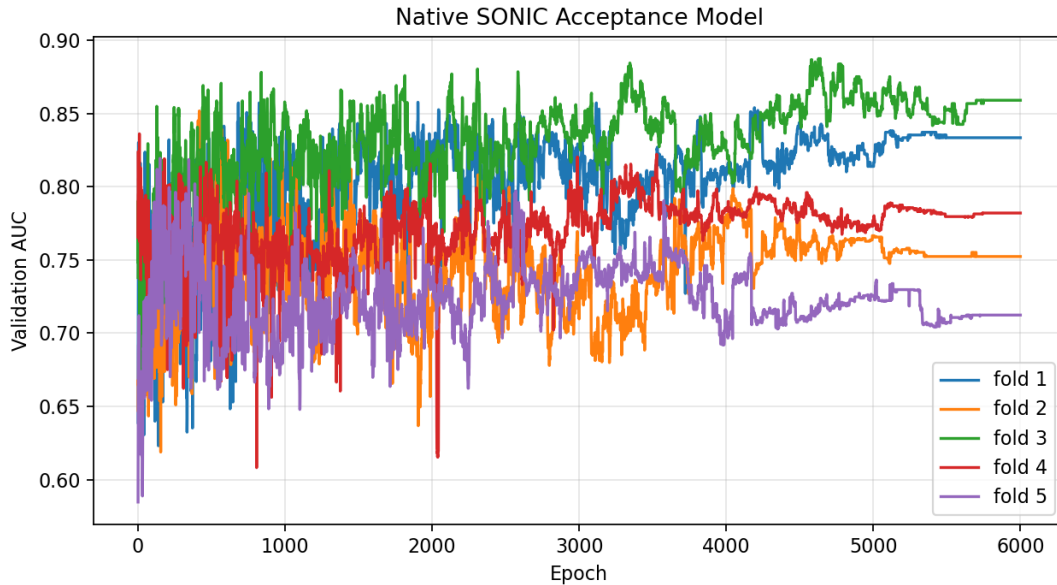


Figure 2: Grouped cross-validation for the broad13 qpos acceptance model. The model is useful for triage, but it is trained on noisy controller labels and should not be interpreted as feasibility certification.

3.5 Visual audit prevents metric-only overclaiming

The learned-selector tracked-camera video audit reports 27 visual passes, 61 warnings, and 16 visual failures over 104 videos. The failures are dominated by crawling collapses and a few upright fall/separation cases. Only one native strict pass is also a visual fail; we exclude that clip from presentation material. The 9 newly evaluated hybrid diagnostic videos have 9/9 visual pass.

4 Discussion

What is submittable. This is defensible as a course final project if framed as controller-in-the-loop curation for a fixed kinematic generator. The most robust claim is: native SONIC exposes a real kinematic-to-dynamic gap; hard gates improve baseline selection; a learned qpos acceptance model predicts controller success better than scalar diagnostics and is useful for abstention/triage; visual audit is necessary to prevent metric-only mistakes.

What is not submittable as a claim. The project does not show MotionBricks fine-tuning, guaranteed physical feasibility, solved crawling, or superiority of the learned selector over the hand-coded gate. The learned rollout is partly same-pool, and the hybrid queue is diagnostic rather than a final win.

Next research direction. The strongest pivot is not another heuristic. It is physical-feedback training: use native tracker outcomes and visual/contact failures to update either the generator distribution or a learned proposer, similar in spirit to controller-feedback alignment methods [5]. Locally, the next small step is to add root-drift/idle sanity to the hybrid selector and rerun a fresh candidate pool. At research scale, the right experiment is to train or adapt a tracking policy or generator with controller outcomes in the loop.

5 Artifacts

- docs/prospective_broad13_2026-05-22.md
- docs/native_acceptance_broad13_2026-05-23.md
- docs/learned_acceptance_prospective_2026-05-23.md
- docs/hybrid_acceptance_queue_2026-05-23.md

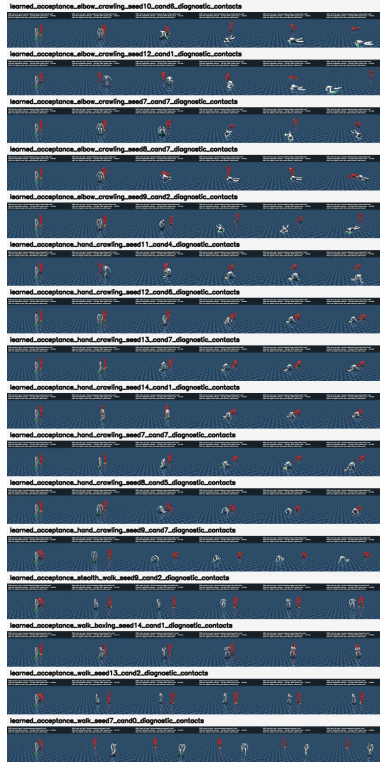


Figure 3: Frame-level visual audit catches failures that numeric tables alone would make too easy to hand-wave. Most visual failures are crawling or fall cases; the final presentation subset excludes visual-fail strict contradictions.

- `results/prospective_native_selection/20260522.broad13/`
- `results/prospective_native_selection/20260523.learned_acceptance_eval/`
- `results/prospective_native_selection/20260523.hybrid_acceptance_queue/`
- `release_assets/cs348k_broad13_results_data_2026-05-22.tar.zst.part-*`
- `release_assets/cs348k_learned_acceptance_eval_2026-05-23.tar.zst`
- `release_assets/cs348k_hybrid_acceptance_queue_2026-05-23.tar.zst`

References

- [1] Zhe Li et al. Smap: Self-supervised motion adaptation for physically plausible humanoid whole-body control, 2025.
- [2] Zhengyi Luo et al. Sonic: Supersizing motion tracking for natural humanoid whole-body control, 2025.
- [3] Junjie Wang et al. Gmt: General motion tracking for humanoid whole-body control, 2025.
- [4] Tingwu Wang, Olivier Dionne, Michael De Ruyter, David Minor, Davis Rempe, Kaifeng Zhao, Mathis Petrovich, Ye Yuan, Chenran Li, Zhengyi Luo, Brian Robison, Xavier Blackwell, Bernardo Antoniazzi, Xue Bin Peng, Yuke Zhu, and Simon Yuen. Motionbricks: Scalable real-time motions with modular latent generative model and smart primitives, 2026.
- [5] Junpeng Yue et al. Rl from physical feedback: Aligning large motion models with humanoid control, 2025.